Concept 1 | Fundamentals of Calculus (Limits and Derivatives)

English Student Edition • Focused formulas and guided practice

Formula opener

Derivative

$$\frac{d}{dx}x^n = nx^{n-1}$$

Product

$$(uv)' = u'v + uv'$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

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Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q001 **Written**

For $f(x) = x^2 + 2x$, find the average rate of change from -1 to 3, and from 3 to $3 + h$.

Q002 Written

Find the average rate of change for the three source Try functions on their stated intervals.

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q003 **Written**

Find the average rate of change for all nine source Exercise functions, including fixed and h-form intervals.

Q004 **Written**

Find the average rate of change of a circle's area as its diameter changes from 10 cm to 18 cm.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q005 Written

Find the source limits by direct substitution and by simplifying the indeterminate fraction before substitution.

Q006 **Written**

Find the three source Try limits, including polynomial and h-quotient forms.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q007 **Written**

Find all nine source Exercise limits, including x-limits and h-limits.

Q008 **Written**

If $\lim_{x\to 3}(x^2 - 5x + k)/(x - 3) = L$ **is finite, find** Lk**.**

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q009 Written

Find the differential coefficient of $f(x) = 3x^2$ at $x = -2$ using the definition.

Q010 **Written**

Find the differential coefficients at the given values for the two source Try functions using the definition.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q011 **Written**

Find the differential coefficients for all six source Exercise functions at their stated values.

Q012 **Written**

A dropped stone has displacement $s(t) = 4.9t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds using the definition.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q013 **Written**

Differentiate $f(x) = x^2 + 3x$ using the definition of the derivative.

Q014 **Written**

Differentiate the three source Try functions using the definition.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q015 **Written**

Differentiate all nine source Exercise functions using the definition.

Q016 **Written**

A regular hexagon has side length x. Write its area and differentiate it using the definition.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q017 **Written**

If $f(x) = 5$ for every x except that it is undefined at x=2, does $\lim_{x \to 2} f(x)$ exist?

Q018 MCQ

The average rate of change from a to b is:

Choose the correct option.

A [f(a)+f(b)]/(b-a)

B [f(b)-f(a)]/(b-a)

C f'(a)

D b-a

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q019 **MCQ**

For $f(x) = x^2 + 2x$ from -1 to 3, the average rate is:

Choose the correct option.

A 2

B 3

C 4

D 8

Q020 MCQ

The average area-rate for diameter 10 to 18 is:

Choose the correct option.

A 7π

B 14π

C 28π

D 56π

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q021 MCQ

The h-form average rate is found by:

Choose the correct option.

A setting $h = 1$

B cancelling the common h

C differentiating first

D squaring h

Q022 MCQ

A direct-substitution limit is valid when:

Choose the correct option.

A the expression is defined

B the denominator is zero

C the numerator is infinite

D h is negative

Classified Practice

Fundamentals of Calculus (Limits and Derivatives)

Q023 MCQ

The 0/0 form requires:

Choose the correct option.

A factor/cancel first

B differentiate twice

C ignore the denominator

D change the base

Q024 MCQ

In the finite-limit challenge, k equals:

Choose the correct option.

A 3

B 5

C 6

D 9

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q025 **MCQ**

For a polynomial, the limit equals:

Choose the correct option.

A its value at the point

B zero

C infinity

D the slope

Q026 MCQ

The definition of the differential coefficient at a is:

Choose the correct option.

A lim [f(a+h)-f(a)]/h

B f(a+h)+f(a)

C [f(a)-f(0)]/a

D h/f(a)

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q027 MCQ

For $f(x) = 3x^2$, $f'(-2)$ is:

Choose the correct option.

A -12

B -6

C 6

D 12

Q028 **MCQ**

The stone's velocity at 10 s is:

Choose the correct option.

A 49

B 98

C 196

D 4.9

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q029 MCQ

After cancelling h, take the limit as:

Choose the correct option.

A $h \to 0$

B $h \to 1$

C $x \to 0$

D $a \to \infty$

Q030 MCQ

The derivative function is:

Choose the correct option.

A the coefficient at one point
B the differential coefficient for every x
C the average rate only
D the original function

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q031 MCQ

The derivative of $x^2 + 3x$ from the definition is:

Choose the correct option.

A x+3
B 2x+3
C 2x
D x^2

Q032 MCQ

The area of a regular hexagon of side x is:

Choose the correct option.

A $3\sqrt{3}x$

B $(3\sqrt{3}/2)x^2$

C $6x^2$

D $\sqrt{3}x$

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Q033 MCQ

A function need not be defined at a for its limit to:

Choose the correct option.

A exist

B be infinite only

C equal zero only

D fail

Q034 **MCQ**

The derivative of a constant function is:

Choose the correct option.

A 0

B 1

C x

D the constant

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Quiz MCQ 1 **Concept Quiz · MCQ**

Find the average rate of change of x^2 from 1 to 3.

Choose the correct option.

A 4
B 1
C 2
D 3

Quiz MCQ 2 **Concept Quiz · MCQ**

Find $\lim_{x\to 2}(x^2 + 1)$.

Choose the correct option.

A 0

B 5

C 2

D 3

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Quiz MCQ 3 Concept Quiz · MCQ

Find $f'(x)$ for $f(x) = x^2$ using the definition.

Choose the correct option.

A 0

B 1

C 2x

D 3

Quiz MCQ 4 **Concept Quiz · MCQ**

If $s(t) = 4.9t^2$, find the velocity at t=5.

Choose the correct option.

A 0

B 1

C 2

D 49

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the average rate of change of $3x + 1$ from 2 to 7.

Quiz WRITTEN 6 **Concept Quiz · Written**

Evaluate $\lim_{h\to 0}(6+2h)$.

Classified Practice **Fundamentals of Calculus (Limits and Derivatives)**

Quiz WRITTEN 7 Concept Quiz · Written

Find the differential coefficient of $2x^2$ at x=3 from the definition.

Quiz WRITTEN 8 **Concept Quiz · Written**

State $\lim_{x \to 2} f(x)$ when f(x)=5 for all x except x=2 where it is undefined.

Answer key

Use the question number shown on each page.

Q001 4; $8+h$

Q002 2; $-5/3$; $h-6$

Q003 Apply $[f(b)-f(a)]/(b-a)$ to each printed item exactly.

Q004 7π cm^2/cm

Q005 -2; 6

Q006 Substitute after removing any common factor; polynomial limits are obtained directly.

Q007 Each limit is obtained by direct substitution or cancellation of the factor causing 0/0.

Q008 6

Q009 $f'(-2)=-12$

Q010 Substitute a into $[f(a+h)-f(a)]/h$, cancel h, and let h tend to zero.

Q011 Use the definition at each specified point and simplify exactly.

Q012 98 m s^{-1}

Q013 $f'(x)=2x+3$

Q014 Expand each $f(x+h)-f(x)$, cancel h and take the limit.

Q015 Apply $f'(x)=\lim_{h\to 0}[f(x+h)-f(x)]/h$ to every printed polynomial.

Q016 $A=\frac{3\sqrt{3}}{2}x^2$, $A'(x)=3\sqrt{3}\,x$

Q017 Yes; the limit is 5.

Q018 B

Q019 C

Q020 A

Q021 B

Q022 A

Q023 A

Q024 C

Q025 A

Q026 A

Q027 A

Q028 B

Q029 A

Q030 B

Q031 B

Q032 B

Q033 A

Q034 A

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 D

Quiz WRITTEN 5 3

Quiz WRITTEN 6 6

Quiz WRITTEN 7 12

Quiz WRITTEN 8 5

Concept 2 | Differentiation and Tangent Lines

English Student Edition • Focused formulas and guided practice

Formula opener

Gradient

$$m = f'(x_1)$$

Tangent

$$y - y_1 = m(x - x_1)$$

Normal

$$m_n m = -1$$

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Classified Practice **Differentiation and Tangent Lines**

Q001 Written

Differentiate the four Warm Up demands: $y = x^2 - 3x + 2$, $y = (2x + 1)^3$, $V = \pi r^2 h$ **with respect to (r), and find** $f'(-1)$ **for** $f(x) = x^3 - 3x + 1$.

Q002 **Written**

Differentiate the six printed Try functions, the two variable-labelled functions $S = \pi r^2$ and $h = vt - \frac{1}{2}gt^2$, and evaluate $f'(2), f'(-3)$ for $f(x) = 4x^3 - 3x^2 + 2$.

Classified Practice Differentiation and Tangent Lines

Q003 Written

Differentiate every source Exercise function $18 polynomial/product forms$**, the six functions with variables indicated in brackets, and evaluate the two values in each of Exercises 3–5.**

Classified Practice **Differentiation and Tangent Lines**

Q004 **Written**

For $f(x) = \sqrt[3]{x^5} + \frac{1}{x^2}$**, find** $f'(1) + f'(-1)$**.**

Q005 **Written**

Find the quadratic $f(x) = ax^2 + bx + c$ **satisfying** $f'(0) = -2, f'(2) = 2, f(1) = 2$.

Classified Practice **Differentiation and Tangent Lines**

Q006 Written

Find the quadratic satisfying $f'(0) = 2, f'(1) = 4, f(2) = 6.$

Q007 **Written**

Find the four quadratics satisfying the four source triples of derivative and function conditions.

Classified Practice **Differentiation and Tangent Lines**

Q008 Written

Find the quadratic (f(x)) satisfying $f'(2) + f'(4) = 0, f(3) = 0, f(1) = 12.$

Q009 **Written**

Find the tangent line to $y = x^2 - 4x + 1$ **at** $(3, -2)$**.**

Classified Practice **Differentiation and Tangent Lines**

Q010 Written

Find the tangent line for the three source functions at their stated points: $y = x^2 + 3$ at $(-2, 7)$, $y = x^2 - 3x + 2$ at ((1,0)), and $y = -x^3 + 2x + 4$ at $(-2, 8)$.

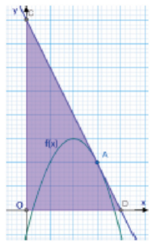
Q011 Written

Find the tangent line for all nine source Exercise functions at their given points.

Classified Practice **Differentiation and Tangent Lines**

Q012 **Written**

For $f(x) = -x^2 + 4x - 1$, the tangent at x-coordinate 3 meets the axes at (D,C). Find the area of triangle (ODC).



Classified Practice **Differentiation and Tangent Lines**

Q013 **Written**

Solve the two source Warm Up demands: tangents from (C(0,2)) to $y = x^2 - 3x + 6$, and tangents to $y = x^3 + 4x^2 - 3$ with gradient 3.

Q014 **Written**

Solve the two source Try demands: tangents from $C(1, -1)$ to $y = x^2 - x + 3$, and tangents to $y = x^3 + 2x - 3$ with gradient 5.

Classified Practice **Differentiation and Tangent Lines**

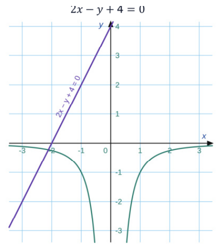
Q015 Written

Solve all six source Exercise demands involving external points or prescribed gradients.

Classified Practice   **Differentiation and Tangent Lines**

Q016 **Written**

For $f(x) = -1/x^2$, find the tangent line parallel to $2x - y + 4 = 0$.



Classified Practice **Differentiation and Tangent Lines**

Q017 MCQ

The derivative of x^n is:

Choose the correct option.

A x^{n-1}

B nx^{n-1}

C $n+x$

D x^n

Q018 MCQ

The derivative of $V = \pi r^2 h$ with respect to r is:

Choose the correct option.

A πh

B $2\pi r h$

C $2\pi r$

D πr^2

Classified Practice **Differentiation and Tangent Lines**

Q019 **MCQ**

For $f(x) = x^3 - 3x + 1$, $f'(-1)$ is:

Choose the correct option.

A -6

B -3

C 0

D 3

Q020 MCQ

The challenge function can be written as:

Choose the correct option.

A $x^{5/3} + x^{-2}$
B $x^5 + x^2$
C $x^{3/5} + x^2$
D $x^{5/3} + x^2$

Classified Practice **Differentiation and Tangent Lines**

Q021 MCQ

For $f(x) = ax^2 + bx + c$, (f'(x)) is:

Choose the correct option.

A $ax + b$

B $2ax + b$

C $2a + b + c$

D $ax^2 + b$

Q022 **MCQ**

The condition $f'(0) = -2$ gives:

Choose the correct option.

A $a = -2$

B $b = -2$

C $c = -2$

D $2a = -2$

Classified Practice **Differentiation and Tangent Lines**

Q023 MCQ

The warm-up quadratic is:

Choose the correct option.

A $x^2 - 2x + 3$

B $x^2 + 2x + 3$

C $2x^2 - x + 3$

D $x^2 - 2x - 3$

Q024 MCQ

The challenge coefficient a is:

Choose the correct option.

A 1

B 2

C 3

D 6

Classified Practice **Differentiation and Tangent Lines**

Q025 MCQ

The tangent gradient at x=a is:

Choose the correct option.

A (f(a))

B (a)

C (f'(a))

D $1/f'(a)$

Q026 MCQ

The tangent to $y = x^2 - 4x + 1$ at $(3, -2)$ is:

Choose the correct option.

A $y = x - 5$

B $y = 2x - 8$

C $y = -2x + 4$

D $y = 3x - 11$

Classified Practice **Differentiation and Tangent Lines**

Q027 MCQ

In the triangle challenge, the x-intercept of the tangent is:

Choose the correct option.

A 2

B 3

C 4

D 8

Q028 MCQ

The area of triangle (ODC) in the challenge is:

Choose the correct option.

A 8

B 12

C 16

D 32

Classified Practice **Differentiation and Tangent Lines**

Q029 MCQ

When the contact point is not given, set it to:

Choose the correct option.

A ((0,0))
B ((a,f(a)))
C ((f(a),a))
D ((1,f’(a)))

Q030 MCQ

The gradient of $2x - y + 4 = 0$ is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice **Differentiation and Tangent Lines**

Q031 MCQ

For $f(x) = -1/x^2$, (f'(x)) is:

Choose the correct option.

A $-2/x^3$
B $2/x^3$
C $1/x^2$
D $-1/x^3$

Q032 MCQ

The parallel tangent in the challenge is:

Choose the correct option.

A $y = 2x + 3$

B $y = 2x - 3$

C $y = -2x - 3$

D $y = x - 2$

Classified Practice **Differentiation and Tangent Lines**

Quiz WRITTEN 1 **Concept Quiz · Written**

Differentiate $3x^3 - 2x$.

Quiz WRITTEN 2 **Concept Quiz · Written**

Find the quadratic with $f'(0) = 1, f'(1) = 3, f(1) = 2.$

Classified Practice **Differentiation and Tangent Lines**

Quiz WRITTEN 3 **Concept Quiz · Written**

Find the tangent to $y = x^2$ at ((2,4)).

Quiz WRITTEN 4 **Concept Quiz · Written**

For $f(x) = -1/x$, find the tangent parallel to $y = x + 2$.

Classified Practice **Differentiation and Tangent Lines**

Quiz MCQ 5 **Concept Quiz · MCQ**

The derivative of a constant is:

Choose the correct option.

A 0

B 1

C x

D the constant

Quiz MCQ 6 **Concept Quiz · MCQ**

For $f(x) = ax^2 + bx + c$, the coefficient equations come from:

Choose the correct option.

A three function values

B two derivatives and one function value

C one derivative only

D the tangent gradient only

Classified Practice **Differentiation and Tangent Lines**

Quiz MCQ 7 Concept Quiz · MCQ

The tangent equation in point-slope form is:

Choose the correct option.

A $y - y_1 = m(x - x_1)$

B $y = mx + c$

C $y + y_1 = m(x + x_1)$

D $x - x_1 = m(y - y_1)$

Quiz MCQ 8 **Concept Quiz · MCQ**

A tangent parallel to $y = 3x - 1$ has gradient:

Choose the correct option.

A -3

B 1/3

C 3

D 1

Answer key

Use the question number shown on each page.

Q001 $2x - 3; 6(2x+1)^2; 2\pi rh; 0$

Q002 $3, -6x+8, 2x+3, 6x^2-8x+6, 6(2x-3)^2, 4x^2+x-1; 2\pi r, v-gt; 36, 126$

Q003 Apply the power rule term-by-term; for a product or power, expand first. The value sets are: Exercise 3 (33,65); Exercise 4 $-24, -3$; Exercise 5 (3,28).

Q004 $\frac{10}{3}$

Q005 $f(x) = x^2 - 2x + 3$

Q006 $f(x) = x^2 + 2x$

Q007 $x^2 - 4x + 8; -2x^2 + 3x + 3; -5x^2 + 26x - 14; -\frac{3}{2}x^2 + 2x + \frac{3}{2}$

Q008 $f(x) = 3x^2 - 18x + 27$

Q009 $y = 2x - 8$

Q010 $y = -4x - 1; y = -x + 1; y = -10x - 12$

Q011 $y = 4x - 5; y = 4x - 1; y = x - 4; y = x - 7; y = -4x + 3; y = 7x - 3; y = 9x - 16; y = -x - 2; y = x + 1$

Q012 16 square units

Q013 $y = -7x + 2, y = x + 2; \quad y = 3x + 15, y = 3x - \frac{95}{27}$

Q014 $y = 5x - 6, y = -3x + 2; \quad y = 5x - 5, y = 5x - 1$

Q015 $y = 6x - 8, y = -2x; y = x + 1, y = -7x + 25; y = -5x + 6, y = 3x - 2; y = 9x - 11, y = 9x + 21; y = -x; y = -2x + 16, y = -2x - 8$

Q016 $y = 2x - 3$

Q017 B

Q018 B

Q019 C

Q020 A

Q021 B

Q022 B

Q023 A

Q024 C

Q025 C

Q026 B

Q027 C

Q028 C

Q029 B

Q030 D

Q031 B

Q032 B

Quiz WRITTEN 1 $9x^2 - 2$

Quiz WRITTEN 2 $x^2 + x$

Quiz WRITTEN 3 $y = 4x - 4$

Quiz WRITTEN 4 $y = x - 4$

Quiz MCQ 5 A

Quiz MCQ 6 B

Quiz MCQ 7 A

Quiz MCQ 8 C

Concept 3 | Applications of Differentiation: Polynomial Functions

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Formula opener

Rate

$$\frac{dy}{dx}$$

Stationary

$$f'(x) = 0$$

Second derivative

$$f''(x) > 0 \Rightarrow \text{minimum}$$

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Classified Practice **Applications of Differentiation: Polynomial Functions**

Q001 **Written**

Find the extreme values and sketch the graphs of the three cubic functions printed in the Warm Up.

Q002 **Written**

Find the extreme values and sketch the three source Try cubics.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q003 **Written**

Find the extreme values and draw the graphs for all nine source cubic functions in Exercise.

Q004 **Written**

For $P(x) = x^3 - 3ax^2 + (3a^2 - 3)x + 4$, $a > 0$, find the flow rates giving the maximum and minimum peak efficiency.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q005 **Written**

On $[-3, 4]$, find the maximum and minimum values of $y = -x^3 + 12x + 5$.

Q006 **Written**

Find the maximum and minimum values of the two source Try cubics on their stated closed intervals.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q007 Written

Find the maximum and minimum values of all six source cubics on their given closed intervals.

Q008 **Written**

For $f(x) = x^3 - 6x$ on $[-3, 3]$, find the absolute deepest dive and highest ascent.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q009 **Written**

If $f(x) = -x^3 + ax + b$ has a local maximum of 9 at $x = 2$, find a, b, and the local minimum.

Q010 **Written**

Solve the two source Try problems: determine the coefficients from a local minimum/maximum and another stationary point.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q011 **Written**

Solve all six source Exercise problems asking for cubic coefficients from extrema.

Q012 **Written**

For $f(x) = x^3 + ax^2 + bx + c$, the graph has a local peak 178 at $x = -2$ and a local trough at $x = 4$. Find a, b, c.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q013 **Written**

If $f(x) = x^3 - 3kx^2 + kx + 2$ has no extreme values, find the range of k.

Q014 **Written**

Determine the parameter ranges in the source Try problems for a cubic to have no extreme values or to have both extremes.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q015 **Written**

Determine the parameter ranges for all six source Exercise cubics according to whether they have extreme values.

Q016 **Written**

For $f(x) = x^3 + 3kx^2 + 12kx + 10$, find the exact k-range that guarantees both a local maximum and minimum.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q017 **Written**

If $f(x) = x^3 - 3kx^2 + (k^2 + 5)x + 3$ is always monotonically increasing, find the range of k.

Q018 **Written**

Find the parameter ranges in the two source Try problems for the cubic to be monotonically increasing.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q019 **Written**

Find the parameter ranges in all six source Exercise problems for monotonic increase.

Q020 **Written**

For $f(x) = \frac{1}{3}x^3 - kx^2 + (k+6)x + 10$, find the k-range for monotonic increase on $x \in [1, \infty)$.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q021 **Written**

Find the number of distinct real roots of $x^3 - 6x^2 + 9x - 3 = 0$.

Q022 **Written**

Find the number of distinct real roots for the three source Try equations.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q023 **Written**

Find the number of distinct real roots for all nine source cubic equations.

Q024 **Written**

Find the number of distinct real roots of $-h^3 + 12h^2 - 36h + 40 = 0$.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q025 **Written**

If $-x^3 + 3x^2 + 9x + a = 0$ has three distinct real roots, find the range of a.

Q026 **Written**

Solve the two source Try parameter problems for three or two distinct real roots.

Classified Practice

Applications of Differentiation: Polynomial Functions

Q027 **Written**

Solve all four source Exercise parameter problems for the required number of distinct roots.

Q028 **Written**

For $f(x) = x^3 - 6x^2 + 9x - 2a^2 + 7a$, find all real a such that $f(x) = 2$ has exactly three distinct real roots.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q029 Written

Prove the source inequalities using an auxiliary cubic and a derivative sign table.

Q030 **Written**

Prove all three source cubic inequalities on their stated domains.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q031 **Written**

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ for every $x \in [-1, 3]$.

Q032 **Written**

Find the extreme values and sketch the two source quartic functions.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q033 **Written**

Find the extreme values and graphs for all six source quartic functions.

Q034 **Written**

For $f(x) = x^4 - 4x^3 + k$, find the exact k-range for exactly two distinct real roots.

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q035 MCQ

At a local maximum, the sign of f' changes:

Choose the correct option.

A $- \rightarrow +$

B $+ \rightarrow -$

C $+ \rightarrow +$

D $- \rightarrow -$

Q036 MCQ

For $P'(x) = 3[(x-a)^2 - 1]$, the local maximum occurs at:

Choose the correct option.

A $a-1$

B a

C $a+1$

D $1-a$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q037 **MCQ**

Absolute extrema on a closed interval require comparing:

Choose the correct option.

A stationary points only

B endpoints only

C stationary points and endpoints

D the y-intercept

Q038 MCQ

The absolute maximum of $x^3 - 6x$ on $[-3, 3]$ is:

Choose the correct option.

A $4\sqrt{2}$

B 9

C 6

D 0

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q039 MCQ

If $f(x) = -x^3 + ax + b$ has a stationary point at 2, then a is:

Choose the correct option.

A 2

B 6

C 12

D -12

Q040 **MCQ**

The derivative roots in the challenge are:

Choose the correct option.

A $-4, 2$

B $-2, 4$

C $2, 4$

D $-3, 3$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q041 MCQ

No two distinct extrema for a cubic means the derivative discriminant is:

Choose the correct option.

A $D > 0$

B $D = 1$

C $D \leq 0$

D $D < 0$ only

Q042 MCQ

The challenge range for both extrema is:

Choose the correct option.

A $0 < k < 4$

B $k < 0$ or $k > 4$

C $-4 < k < 0$

D $k \leq 4$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q043 MCQ

For a positive-leading cubic to be monotone increasing, its derivative must be:

Choose the correct option.

A negative everywhere
B non-negative everywhere
C zero everywhere
D linear

Q044 MCQ

The challenge domain in 7-13 is:

Choose the correct option.

A $(-\infty, \infty)$

B $[1, \infty)$

C $[-1, 1]$

D $(0, 1)$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q045 **MCQ**

Three distinct roots occur when the x-axis lies:

Choose the correct option.

A above both extrema
B between the local maximum and minimum
C at an endpoint only
D at the local maximum only

Q046 MCQ

The 7-14 challenge has how many distinct real roots?

Choose the correct option.

A 0

B 1

C 2

D 3

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q047 MCQ

Three intersections with $y = a$ require the level to lie:

Choose the correct option.

A strictly between the extrema

B above the maximum

C below the minimum

D at an extremum

Q048 MCQ

The 7-15 warm-up range is:

Choose the correct option.

A $-27 < a < 5$
B $-5 < a < 27$
C $a \leq -27$
D $a \geq 5$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q049 MCQ

To prove LHS $\geq$ RHS, define:

Choose the correct option.

A $f = \text{LHS} + \text{RHS}$

B $f = \text{LHS} - \text{RHS}$

C $f = \text{RHS} - 1$

D $f = f'$

Q050 MCQ

The largest k in the challenge is:

Choose the correct option.

A $2-4\sqrt{2}$

B $4\sqrt{2}-2$

C $2+4\sqrt{2}$

D $-2-4\sqrt{2}$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q051 MCQ

A repeated root of a derivative may be:

Choose the correct option.

A always a local maximum
B always a local minimum
C stationary but not an extreme
D impossible

Q052 **MCQ**

Exactly two distinct roots for $x^4 - 4x^3 + k = 0$ require:

Choose the correct option.

A $k = 27$

B $k < 27$

C $k > 27$

D $k \leq 0$ only

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q053 MCQ

A local minimum is identified by the change:

Choose the correct option.

A $+ \to -$

B $- \to +$

C $+ \to +$

D $- \to -$

Q054 MCQ

For $y = x^3 - 6x^2 + 16$, the critical x-values are:

Choose the correct option.

A $0, 4$

B $1, 3$

C $-2, 2$

D $2, 4$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q055 **MCQ**

The derivative sign table is used to determine:

Choose the correct option.

A only the y-intercept

B increase/decrease and extrema

C the constant term

D the domain

Q056 MCQ

On a closed interval, an endpoint can be:

Choose the correct option.

A ignored
B an absolute extreme
C a derivative root only
D never a minimum

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q057 MCQ

For $y = -x^3 + 12x + 5$, the stationary x-values are:

Choose the correct option.

A $-2, 2$

B $-1, 1$

C $0, 4$

D $2, 4$

Q058 **MCQ**

The absolute minimum of $x^3 - 6x$ on $[-3, 3]$ is at:

Choose the correct option.

A $x = -3$
B $x = -\sqrt{2}$
C $x = \sqrt{2}$
D $x = 3$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q059 MCQ

To determine a cubic from an extreme value, use both $f(a) = M$ and:

Choose the correct option.

A $f(a) = 0$

B $f'(a) = 0$

C $f''(a) = 0$

D $a = 0$

Q060 MCQ

In the warm-up $f(x) = -x^3 + 12x - 7$, the local minimum occurs at:

Choose the correct option.

A $x = -2$

B $x = 0$

C $x = 2$

D $x = 7$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q061 **MCQ**

For $f(x) = x^3 - 3kx^2 + kx + 2$, no two distinct stationary roots requires:

Choose the correct option.

A $0 \leq k \leq 1/3$

B $k > 1/3$

C $k < 0$

D $k \geq 1/3$

Q062 **MCQ**

Two distinct stationary points occur when the derivative discriminant is:

Choose the correct option.

A $D < 0$

B $D = 0$

C $D > 0$

D $D = 1$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q063 MCQ

For a positive-leading quadratic derivative, $D \leq 0$ implies it is:

Choose the correct option.

A non-negative for all x

B negative for all x

C linear

D undefined

Q064 MCQ

In the 7-13 challenge, monotonicity is required on:

Choose the correct option.

A $[1, \infty)$

B $[-1, 1]$

C $(-infty, 0)$

D $[0, 1]$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q065 MCQ

A cubic with both stationary values above the x-axis has:

Choose the correct option.

A three roots

B one distinct real root

C no real roots

D two repeated roots

Q066 MCQ

To count cubic roots, compare the extrema with:

Choose the correct option.

A the y-axis
B the x-axis
C the derivative axis
D the domain only

Classified Practice **Applications of Differentiation: Polynomial Functions**

Q067 MCQ

At an extremal parameter level, a cubic and a horizontal line have:

Choose the correct option.

A one distinct intersection
B two distinct intersections
C three distinct intersections
D no intersection

Q068 **MCQ**

For $x^4 - 4x^3 + k$, the global minimum occurs at:

Choose the correct option.

A $x = 0$

B $x = 1$

C $x = 3$

D $x = 4$

Classified Practice **Applications of Differentiation: Polynomial Functions**

Quiz WRITTEN 1 **Concept Quiz · Written**

Find the local maximum and minimum of $x^3 - 3x^2 + 1$.

Quiz WRITTEN 2 **Concept Quiz · Written**

Find a, b **if** $f(x) = -x^3 + ax + b$ **has a local maximum 4 at x=1.**

Classified Practice **Applications of Differentiation: Polynomial Functions**

Quiz WRITTEN 3 **Concept Quiz · Written**

Find the k-range for $x^3 - 3kx^2 + kx$ to have no two distinct stationary points.

Quiz WRITTEN 4 **Concept Quiz · Written**

Find the largest k such that $x^3 - 3x^2 - 3x + k + 3 \leq 0$ **on** $[-1, 3]$**.**

Classified Practice **Applications of Differentiation: Polynomial Functions**

Quiz MCQ 5 Concept Quiz · MCQ

At a local minimum, f' changes:

Choose the correct option.

A $+ \to -$

B $- \to +$

C $+ \to +$

D $- \to -$

Quiz MCQ 6 **Concept Quiz · MCQ**

A cubic with a positive-leading derivative and $D < 0$ is:

Choose the correct option.

A always increasing

B always decreasing

C constant

D undefined

Classified Practice **Applications of Differentiation: Polynomial Functions**

Quiz MCQ 7 Concept Quiz · MCQ

The 7-15 challenge requires the level to satisfy:

Choose the correct option.

A $t < 0$

B $0 < t < 4$

C $t > 4$

D $t = 4$

Quiz MCQ 8 **Concept Quiz · MCQ**

The global minimum of $x^4 - 4x^3 + k$ is:

Choose the correct option.

A $k - 27$
B $k + 27$
C $k - 3$
D $27 - k$

Answer key

Use the question number shown on each page.

Q001 Use the derivative sign table: for x^3-6x^2+16, local maximum 16 at $x=0$ and local minimum -16 at $x=4$; classify the other two from their stationary points.

Q002 Apply the same derivative sign-table test; a stationary point where the sign of f' does not change is not an extreme value.

Q003 Differentiate each polynomial, solve the quadratic derivative, use the sign table, and evaluate f at every genuine stationary point.

Q004 Local maximum at $x=a-1$; local minimum at $x=a+1$.

Q005 Maximum 21 at $x=2$; minimum -11 at $x=-2$.

Q006 Evaluate both endpoints and every stationary point inside the interval; the greatest value is the absolute maximum and the least is the absolute minimum.

Q007 For each function, compare the endpoint values with the values at all critical x-values lying in the interval.

Q008 Absolute minimum -9 at $x=-3$; absolute maximum 9 at $x=3$.

Q009 $a=12, b=-7$; local minimum -23 at $x=-2$.

Q010 Use $f'(a)=0, f(a)=M$, and the second stationary condition to form three simultaneous equations in the coefficients.

Q011 Each set is determined by two derivative roots/conditions and one extreme-value condition; solve the resulting simultaneous equations.

Q012 $a=-3,\ b=-24,\ c=150.$

Q013 $0\le k\le \frac{1}{3}.$

Q014 Use $D\le 0$ for no extremes and $D>0$ for two distinct stationary points.

Q015 Apply the derivative-discriminant criterion separately to each printed cubic.

Q016 $k<0 or k>4.$

Q017 $-\frac{\sqrt{10}}{2}\le k\le \frac{\sqrt{10}}{2}.$

Q018 Set the derivative discriminant $D\le 0$, with the derivative leading coefficient positive.

Q019 Differentiate each cubic and impose that its quadratic derivative never becomes negative.

Q020 $k\le 3.$

Q021 Three distinct real roots.

Q022 Use the cubic graph: count x-axis intersections from the stationary points and end behavior.

Q023 Each answer is obtained from the number of x-axis intersections of its corresponding cubic graph.

Q024 One distinct real root.

Q025 $-27<a<5.$

Q026 Rewrite each equation as $g(x)=a$, then compare the parameter line with the local maximum and minimum of g.

Q027 Use the local extrema of the parameter-free cubic and count intersections with the horizontal parameter line.

Q028 $\frac{7-\sqrt{65}}{4}<a<\frac{7-\sqrt{33}}{4} or \frac{7+\sqrt{33}}{4}<a<\frac{7+\sqrt{65}}{4}.$

Q029 Set $f(x)=\text{LHS}-\text{RHS}$, prove $f(x)\ge 0$ on the stated domain, and identify its minimum.

Q030 Move the right-hand side to the left, then show the resulting auxiliary cubic has non-negative minimum on the domain.

Q031 $k\le 2-4\sqrt{2}$; the largest value is $2-4\sqrt{2}$.

Q032 For $3x^4+4x^3-12x^2$: local maximum 0 at $x=0$, local minima -32 at $x=-2$ and -5 at $x=1$. For $-x^4-4x^3-8$: maximum 19 at $x=-3$; $x=0$ is stationary but not an extreme.

Q033 Differentiate each quartic, construct the derivative sign table, and evaluate each genuine turning point.

Q034 $k<27.$

Q035 B

Q036 A

Q037 C

Q038 B

Q039 C

Q040 B

Q041 C

Q042 B

Q043 B

Q044 B

Q045 B

Q046 B

Q047 A

Q048 A

Q049 B

Q050 A

Q051 C

Q052 B

Q053 B

Q054 A

Q055 B

Q056 B

Q057 A

Q058 A

Q059 B

Q060 A

Q061 A

Q062 C

Q063 A

Q064 A

Q065 B

Q066 B

Q067 B

Q068 C

Quiz WRITTEN 1 Maximum 1 at $x = 0$; minimum -3 at $x = 2$

Quiz WRITTEN 2 $a = 3,\ b = 2$

Quiz WRITTEN 3 $-\frac{\sqrt{10}}{2} \le k \le \frac{\sqrt{10}}{2}$

Quiz WRITTEN 4 $2 - 4\sqrt{2}$

Quiz MCQ 5 B

Quiz MCQ 6 A

Quiz MCQ 7 B

Quiz MCQ 8 A

Concept 4 | Integration: Definite Integrals and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x)dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)|dx$$

Eng Eslam Ahmed | 01120009622

Classified Practice **Integration: Definite Integrals and Applications**

Q001 **Written**

Find the two indefinite integrals in the Warm Up.

Q002 **Written**

Find the six indefinite integrals in the source Try section, expanding a squared bracket first when needed.

Classified Practice Integration: Definite Integrals and Applications

Q003 Written

Solve the first ten source indefinite-integral exercises (constants, linear and quadratic polynomials).

Q004 Written

Solve the remaining eight source integrals, including products and squared brackets.

Classified Practice **Integration: Definite Integrals and Applications**

Q005 **Written**

Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.

Q006 **Written**

Find the two source Try functions from their derivatives and one given value of F.

Classified Practice **Integration: Definite Integrals and Applications**

Q007 **Written**

Find $F(x)$ for all six source Exercise conditions.

Q008 **Written**

Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.

Classified Practice **Integration: Definite Integrals and Applications**

Q009 **Written**

Solve the source Try tangent-gradient problem through $(1, 1)$.

Q010 Written

Solve the three source tangent-gradient exercises, finding each function from its point and gradient.

Classified Practice **Integration: Definite Integrals and Applications**

Q011 **Written**

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Q012 **Written**

Evaluate the six source Try definite integrals, simplifying the integrand before applying the limits.

Classified Practice **Integration: Definite Integrals and Applications**

Q013 **Written**

Evaluate the first twelve source Exercise definite integrals, including reversed and equal limits.

Q014 **Written**

Evaluate the remaining twelve source Exercise integrals, including sums, differences and products.

Classified Practice **Integration: Definite Integrals and Applications**

Q015 **Written**

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Q016 **Written**

Solve the four source Try definite-integral transformations.

Classified Practice **Integration: Definite Integrals and Applications**

Q017 **Written**

Solve the first six source Exercise questions using interval splitting and reversal.

Q018 **Written**

Solve the final six source Exercise questions using the factorized definite-integral identity.

Classified Practice **Integration: Definite Integrals and Applications**

Q019 **Written**

Find $f(x)$ from the source equation $f(x) = x^2 + \int_0^1 f(t)\,dt$.

Q020 **Written**

Solve the two source Try integral equations for $f(x)$.

Classified Practice **Integration: Definite Integrals and Applications**

Q021 **Written**

Find $f(x)$ for all five source integral equations.

Q022 **Written**

Differentiate $\int_3^x (2t^2 + 3t - 5)\,dt$ **with respect to** x**.**

Classified Practice **Integration: Definite Integrals and Applications**

Q023 **Written**

Solve the two source Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Q024 **Written**

Differentiate the three source expressions with respect to x.

Classified Practice **Integration: Definite Integrals and Applications**

Q025 **Written**

Find $f(x)$ and a for the three source equations with a variable upper limit.

Q026 MCQ

The constant of integration is required because:

Choose the correct option.

A every primitive is unique
B primitives differ by a constant
C the integrand is constant
D the limits are equal

Classified Practice **Integration: Definite Integrals and Applications**

Q027 **MCQ**

The integral of x^n for $n \neq -1$ is:

Choose the correct option.

A $nx^{n-1} + C$

B $x^{n+1}/(n+1) + C$

C $x^n/n + C$

D $x^{n-1}/(n-1) + C$

Q028 MCQ

Before integrating $(x-1)^2$, first:

Choose the correct option.

A differentiate it

B expand it

C set $x = 0$

D add limits

Classified Practice **Integration: Definite Integrals and Applications**

Q029 MCQ

The integral of $3x^2 - 2x + 1$ is:

Choose the correct option.

A $x^3 - x^2 + x + C$

B $3x^3 - 2x^2 + x + C$

C $x^3 - 2x + 1 + C$

D $6x - 2 + C$

Q030 MCQ

To find $F(x)$ from $F'(x)$, you should first:

Choose the correct option.

A differentiate again

B integrate $F'(x)$

C set $F = 0$

D ignore the initial value

Classified Practice **Integration: Definite Integrals and Applications**

Q031 **MCQ**

The condition $F(1) = 7$ determines:

Choose the correct option.

A the derivative

B the domain

C the constant C

D the leading coefficient

Q032 MCQ

If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:

Choose the correct option.

A $3x^2 - 2x + 1$

B $6x^2 - 2x + 1$

C $3x^2 - 2x$

D $6x - 2 + 1$

Classified Practice **Integration: Definite Integrals and Applications**

Q033 MCQ

The gradient of the tangent at (x, y) is:

Choose the correct option.

A $f(x)$
B $f'(x)$
C $f''(x)$
D $x + y$

Q034 MCQ

A point (x_0, y_0) on $y = f(x)$ gives:

Choose the correct option.

A $f'(x_0) = y_0$

B $f(x_0) = y_0$

C $f(0) = x_0$

D $C = 0$

Classified Practice **Integration: Definite Integrals and Applications**

Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Choose the correct option.

A $3x^3 - 2x^2 + x$

B $x^3 - x^2 + x + C$

C $6x - 2$

D $x^3 - 2x + 1$

Q036 MCQ

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Choose the correct option.

A $F(a) - F(b)$

B $F(b) - F(a)$

C $F(a) + F(b)$

D 0 always

Classified Practice **Integration: Definite Integrals and Applications**

Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:

Choose the correct option.

A 1
B a
C 0
D $f(a)$

Q038 MCQ

A reversed definite integral satisfies:

Choose the correct option.

A $\int_a^b f = \int_b^a f$

B $\int_a^b f = -\int_b^a f$

C both are zero

D only if $f = 0$

Classified Practice **Integration: Definite Integrals and Applications**

Q039 MCQ

To evaluate a definite integral of a product, first:

Choose the correct option.

A integrate each factor separately
B expand the product
C differentiate the product
D change the limits

Q040 MCQ

Adjacent integrals with the same integrand can be:

Choose the correct option.

A multiplied

B combined over the joined interval

C set equal to zero

D reversed only

Classified Practice **Integration: Definite Integrals and Applications**

Q041 **MCQ**

To use $\int_{\alpha}^{\beta}(x-\alpha)(x-\beta)dx$, the factors must have:

Choose the correct option.

A different roots

B the displayed roots (a,b)

C zero coefficients

D equal limits

Q042 **MCQ**

The identity for the factored integral is:

Choose the correct option.

A $(\beta - \alpha)^2/6$

B $-(\beta - \alpha)^3/6$

C $(\beta + \alpha)^3/6$

D 0

Classified Practice **Integration: Definite Integrals and Applications**

Q043 MCQ

A minus sign before an integral can be removed by:

Choose the correct option.

A squaring the integrand

B reversing its limits

C adding C

D differentiating

Q044 MCQ

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Choose the correct option.

A a function of x

B a constant

C always zero

D a derivative

Classified Practice **Integration: Definite Integrals and Applications**

Q045 **MCQ**

The best first step for an integral equation is to let:

Choose the correct option.

A $A = f(x)$

B $A = \int_a^b f(t)dt$

C $A = x$

D $A = C$ only

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:

Choose the correct option.

A 1/3

B $-1/3$

C 0

D 1

Classified Practice **Integration: Definite Integrals and Applications**

Q047 **MCQ**

$\frac{d}{dx}\int_a^x f(t)dt$ **equals:**

Choose the correct option.

A $f(a)$

B $f(x)$

C $F(x)$

D 0

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:

Choose the correct option.

A the lower limit

B the upper-limit variable

C C

D zero

Classified Practice **Integration: Definite Integrals and Applications**

Q049 **MCQ**

If $\int_a^a f(t)dt = 0$, this can determine:

Choose the correct option.

A the derivative

B the unknown lower limit a

C the integrand degree

D the upper limit

Q050 MCQ

For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:

Choose the correct option.

A $2x^2 + 3x - 5$

B $2x^3 + 3x^2 - 5x$

C $3x^2 + 3x - 5$

D 0

Classified Practice **Integration: Definite Integrals and Applications**

Quiz WRITTEN 1 **Concept Quiz · Written**

Find $int(4x^3 - 6x + 2)dx$.

Quiz WRITTEN 2 **Concept Quiz · Written**

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Classified Practice **Integration: Definite Integrals and Applications**

Quiz WRITTEN 3 **Concept Quiz · Written**

Evaluate $\int_{-1}^{2}(3x^2 - 4x + 1)dx$.

Quiz WRITTEN 4 **Concept Quiz · Written**

Find $f(x)$ **if** $f(x) = x^2 + \int_0^1 f(t)dt$.

Classified Practice **Integration: Definite Integrals and Applications**

Quiz MCQ 5 **Concept Quiz · MCQ**

The derivative of $\int_0^x (3t^2+1)dt$ is:

Choose the correct option.

A $3x^2+1$

B x^3+x

C $6x$

D 0

Quiz MCQ 6 **Concept Quiz · MCQ**

A primitive of $5x^4$ is:

Choose the correct option.

A $x^5 + C$

B $5x^5 + C$

C $20x^3 + C$

D $x^4/5 + C$

Classified Practice **Integration: Definite Integrals and Applications**

Quiz MCQ 7 **Concept Quiz · MCQ**

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Choose the correct option.

A $2x^2 + 2x + 3$
B $4x^2 + 2x + 3$
C $2x^2 + 2$
D $4x + 5$

Quiz MCQ 8 **Concept Quiz · MCQ**

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ **equals:**

Choose the correct option.

A $\int_1^5 f(x)dx$
B $\int_1^3 f(x)dx$
C $\int_3^5 f(x)dx$
D 0

Answer key

Use the question number shown on each page.

Q001 Integrate term by term and add C to each primitive.

Q002 Expand products or powers, collect like powers, then integrate every term.

Q003 Apply linearity and the power rule; retain the constant of integration.

Q004 Expand each product or square, simplify, and integrate term by term.

Q005 $F(x) = x^3 + 4x^2 - x + 3$.

Q006 Integrate the derivative and use the given point to determine C.

Q007 For each item, integrate $F'(x)$, then impose the stated value of F.

Q008 $f(x) = x^3 - x^2 + x - 19$.

Q009 Integrate the given gradient and use the point on the curve to determine the constant.

Q010 Integrate the polynomial gradient and apply the given point condition in each case.

Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.

Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.

Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.

Q014 Use linearity, expand products, then apply the endpoint-substitution rule.

Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.

Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.

Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits.

Q018 Scale to $\int_a^b (x - a)(x - b)\,dx = -\frac{(b-a)^3}{6}$.

Q019 Let $A = \int_0^{\,1} f(t)\,dt$; then $A = frac13 + A$, so $A = -frac13$ and $f(x) = x^2 - frac13$.

Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.

Q021 Let the definite integral be A, substitute the expression for $f(t)$, solve for A, then recover $f(x)$.

Q022 $2x^2 + 3x - 5$.

Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a.

Q024 Apply the variable-limit Fundamental Theorem directly to each integrand.

Q025 Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a.

Q026 B

Q027 B

Q028 B

Q029 A

Q030 B

Q031 C

Q032 A

Q033 B

Q034 B

Q035 B

Q036 B

Q037 C

Q038 B

Q039 B

Q040 B

Q041 B

Q042 B

Q043 B

Q044 B

Q045 B

Q046 B

Q047 B

Q048 B

Q049 B

Q050 A

Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$

Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$

Quiz WRITTEN 3 6

Quiz WRITTEN 4 $f(x) = x^2 - 1/3$

Quiz MCQ 5 A

Quiz MCQ 6 A

Quiz MCQ 7 A

Quiz MCQ 8 A

Concept 5 | Applications of Integration to Geometry

English Student Edition • Focused formulas and guided practice

Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom}) dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \le x \le b$$

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Classified Practice Applications of Integration to Geometry

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x-axis, $x = -2$, and $x = 1$.

Q002 **Written**

Find the area bounded by $y = x^2 + x - 6$, the x-axis, $x = -2$, and $x = 1$.

Classified Practice **Applications of Integration to Geometry**

Q003 **Written**

Find the area bounded by $y = -x^2 + 4$ and the x-axis.

Q004 **Written**

Find the area under $y = -x^2 - 2x + 3$ and above the x-axis.

Classified Practice Applications of Integration to Geometry

Q005 Written

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.

Q006 **Written**

Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.

Classified Practice **Applications of Integration to Geometry**

Q007 **Written**

Find the area between $y = -x^2 + 4x$ and $y = x$.

Q008 Written

Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \leq x \leq 3$.

Classified Practice Applications of Integration to Geometry

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x-axis.

Q010 **Written**

Find the area bounded by $y = -x^3 + 4x$ and the x-axis.

Classified Practice **Applications of Integration to Geometry**

Q011 **Written**

Solve the source cubic area exercises by finding every x-intercept and adding the absolute signed pieces.

Q012 **Written**

Evaluate $\int_{-2}^{1} |x+1| \, dx$.

Classified Practice **Applications of Integration to Geometry**

Q013 **Written**

Evaluate $\int_0^3 |x^2 - 4|\, dx$.

Q014 **Written**

Evaluate $\int_0^2 |x^2 + 3x - 4|\, dx$.

Classified Practice **Applications of Integration to Geometry**

Q015 **Written**

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Q016 **Written**

Find the area bounded by $y = x^3 + x^2 - 3x + 6$ **and its tangent at** $(1, 5)$.

Classified Practice **Applications of Integration to Geometry**

Q017 **Written**

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Q018 **Written**

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x-axis, find k.

Classified Practice Applications of Integration to Geometry

Q019 Written

Solve the source bisection exercises for the given parabolas and lines, using the required area ratio.

Q020 **MCQ**

When the graph is below the x-axis, area is:

Choose the correct option.

A the signed integral

B the negative of the integral

C always zero

D undefined

Classified Practice

Applications of Integration to Geometry

Q021 **MCQ**

A bounded area under a curve is found by first checking:

Choose the correct option.

A the graph and sign

B only the y-intercept

C the font size

D the derivative only

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:

Choose the correct option.

A 8/3

B 16/3

C 32/3

D 64/3

Classified Practice Applications of Integration to Geometry

Q023 MCQ

Between two curves, the integrand is:

Choose the correct option.

A lower-upper

B upper-lower

C product

D derivative

Q024 MCQ

If curves cross inside the interval, area must be:

Choose the correct option.

A split at the crossing

B ignored

C doubled always

D negative

Classified Practice **Applications of Integration to Geometry**

Q025 MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A 16/3

B 32/3

C 64/3

D 9/2

Q026 MCQ

For cubic area, the sign changes at:

Choose the correct option.

A every turning point

B each x-intercept

C the y-intercept only

D the midpoint

Classified Practice **Applications of Integration to Geometry**

Q027 **MCQ**

The area for $y = -x^3 + 4x$ is:

Choose the correct option.

A 4

B 6

C 8

D 12

Q028 MCQ

A negative lobe contributes:

Choose the correct option.

A negative area

B positive absolute area

C no area

D a derivative

Classified Practice **Applications of Integration to Geometry**

Q029 **MCQ**

For an absolute-value integral, first locate:

Choose the correct option.

A the roots

B the vertex only

C the y-axis

D the tangent

Q030 MCQ

The value of $\int_0^2 |x^2+3x-4|dx$ is:

Choose the correct option.

A 3

B 4

C 5

D 6

Classified Practice **Applications of Integration to Geometry**

Q031 MCQ

If f changes sign at c, split the integral:

Choose the correct option.

A at c

B at 0 only

C never

D after doubling

Q032 MCQ

A tangent line at x=a has slope:

Choose the correct option.

A f(a)

B f'(a)

C a^2

D the area

Classified Practice **Applications of Integration to Geometry**

Q033 **MCQ**

For two tangent lines, split at their:

Choose the correct option.

A intersection x-coordinate
B y-intercept only
C vertex only
D origin

Q034 MCQ

The area in the source parabola-tangent Warm Up is:

Choose the correct option.

A 1/2

B 1

C 3/2

D 3

Classified Practice Applications of Integration to Geometry

Q035 MCQ

Bisection means each part has:

Choose the correct option.

A the same area
B the same slope
C the same root
D zero area

Q036 MCQ

For $y=2x-x^2$, the total area above the x-axis is:

Choose the correct option.

A 1

B 2

C 3

D 4

Classified Practice **Applications of Integration to Geometry**

Q037 **MCQ**

The line y=kx must satisfy the stated:

Choose the correct option.

A area ratio

B font ratio

C root count

D period

Q038 MCQ

In a bisection problem, the cut-off area is:

Choose the correct option.

A the total area

B half the total area

C twice the total area

D zero

Classified Practice **Applications of Integration to Geometry**

Quiz MCQ 1 Concept Quiz · MCQ

The area under $y = -x^2 + 4$ and above the x-axis is:

Choose the correct option.

A 16/3

B 8

C 32/3

D 64/3

Quiz MCQ 2 **Concept Quiz · MCQ**

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Choose the correct option.

A 16/3

B 32/3

C 64/3

D 9/2

Classified Practice **Applications of Integration to Geometry**

Quiz MCQ 3 **Concept Quiz · MCQ**

Evaluate $\int_{-2}^{1} |x+1|\, dx$:

Choose the correct option.

A 2

B 5/2

C 3

D 7/2

Quiz MCQ 4 **Concept Quiz · MCQ**

The area enclosed by $y = x^2$ and tangents through $(3, 8)$ is:

Choose the correct option.

A 1/3
B 1/2
C 2/3
D 4/3

Classified Practice **Applications of Integration to Geometry**

Quiz WRITTEN 5 Concept Quiz · Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x-axis.

Quiz WRITTEN 6 **Concept Quiz · Written**

Evaluate $\int_0^2 |x^2 + 3x - 4|\, dx$.

Classified Practice **Applications of Integration to Geometry**

Quiz WRITTEN 7 Concept Quiz · Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Quiz WRITTEN 8 **Concept Quiz · Written**

If $y = kx$ bisects the area under $y = 2x - x^2$, find k.

Answer key

Use the question number shown on each page.

Q001 9
Q002 $\frac{33}{2}$
Q003 $\frac{32}{3}$
Q004 $\frac{32}{3}$
Q005 $\frac{32}{3}$
Q006 $\frac{64}{3}$
Q007 $\frac{9}{2}$
Q008 30
Q009 $\frac{37}{12}$
Q010 8
Q011 Use the numbered solution for each source cubic.
Q012 $\frac{5}{2}$
Q013 $\frac{23}{3}$
Q014 5
Q015 $\frac{3}{2}$
Q016 $\frac{64}{3}$
Q017 $\frac{2}{3}$
Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$
Q019 Use the numbered solution for each source ratio.
Q020 B
Q021 A
Q022 C
Q023 B
Q024 A
Q025 B
Q026 B
Q027 C
Q028 B
Q029 A
Q030 C
Q031 A
Q032 B
Q033 A
Q034 C
Q035 A
Q036 C
Q037 A
Q038 B
Quiz MCQ 1 C
Quiz MCQ 2 B
Quiz MCQ 3 B
Quiz MCQ 4 C
Quiz WRITTEN 5 $\frac{37}{12}$
Quiz WRITTEN 6 5
Quiz WRITTEN 7 $\frac{3}{2}$
Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$